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International sojourn experience and personality development: Selection and socialization effects of studying abroad and the Big Five

Esther Niehoff *, Linn Petersdotter, Philipp Alexander Freund

Leuphana University of Lüneburg, Germany

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ABSTRACT

As part of a multi-study project, this test-retest study seeks to identify the relations between studying abroad and a sojourner's personality as measured by the Big Five personality traits. It thereby attempts to answer the questions of who chooses to study abroad and how study abroad changes personality. A total of 221 students from a German university were tracked over the course of a semester, with the Big Five being obtained via a German version of the Big Five Inventory (Lang, Lüdtke, & Asendorpf, 2001) both at the beginning and at the end. The share of 93 students who studied abroad were found to rate higher in agreeableness and openness prior to the international experience than their fellow students who did not sojourn. In turn, sojourning evoked increases in both extraversion and agreeableness and a decrease in neuroticism. Upon inclusion of interaction terms of initial Big Five levels and study abroad status, positive main effects of study abroad and negative interaction effects for both agreeableness and conscientiousness could be observed.

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1. Introduction

Consequences of expanding globalization and growing internationalization are noticeable in almost every aspect of life today and make no exception for the field of academic education. In the past decades, the amount of students with international sojourn experience has risen continually and there appears to be no end to this trend. Currently, about 37% of Germany's students enrolled in higher semesters have completed a study-related visit abroad. A total of 29% even spent at least three months abroad (Deutscher Akademischer Austauschdienst, 2015). Though study abroad programs seek to provide valuable academic experiences, students have indicated that their primary reasons for studying abroad were less related to gaining methodological and technical expertise but rather to experiencing personal development and growth (Heublein, Hutzsch, Schreiber, & Sommer, 2011).

As a result of this development, recent research has focused on psychological conditions and consequences of foreign exposure (Jokela, 2009; Ying, 2002). However, only few studies have specifically examined the ways and extent to which personality changes as a result of study-related international sojourn experiences (e.g., Zimmermann & Neyer, 2013). The present study therefore seeks to further contribute to this debate by investigating the relation between studying abroad and the Big Five personality dimensions.

1.1. Stability and change in personality

For decades, personality researchers have classified personality traits as stable patterns, which characterize individuals in a distinguishing way. The well-established Five Factor model (extraversion, agreeableness, conscientiousness, neuroticism, and openness) is widely accepted for its high reliability and temporal stability (McCrae & Costa, 2008). However, in recent years, more and more longitudinal studies have shown that personality is subject to change to a larger degree than widely assumed (e.g., Specht et al., 2014; Lucas & Donnellan, 2011).

Previous research has especially focused on mean-level change of personality scores across time (Roberts, Walton, & Viechtbauer, 2006). Various studies were able to identify this kind of normative change across the life course (Lucas & Donnellan, 2011; Specht, Egloff, & Schmukle, 2011; Robins, Fraley, Roberts, & Trzesniewski, 2001). A meta-analysis by Roberts et al. (2006) indicated that conscientiousness, agreeableness, and emotional stability increase with age, whereas for extraversion and openness change patterns seem to depend on the specific facet or phase of life under study (Roberts et al., 2006). Up to the present day researchers have not been able to reach a consensus regarding a causal explanation for this empirically well-supported plasticity among the Big Five (Specht et al., 2014).

1.2. Types of change in personality

Personality change can occur in consequence of intrinsic, i.e. genetic factors, as well as extrinsic, i.e. environmental causes (Specht et al.,

* Corresponding author at: Fritz-Erler Straße 30, 28279 Bremen, Germany.
E-mail address: esther.niehoff@gmx.de (E. Niehoff).

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2014). Recently, results of an increasing number of studies indicate that a one-sided perspective, whether essentialist or contextualist, is insufficient to explain personality development comprehensively (Roberts, Wood, & Caspi, 2008; Specht et al., 2011). For this reason, a transactional approach combining both perspectives was introduced: the model of person-environment transactions (Roberts et al., 2008). The model postulates a mutual influence between personality and environment.

Regarding the direction of the reciprocal influence between an individual's personality and its environment, it is possible to differentiate between two types of effects: selection effects and socialization effects. Selection effects refer to the mechanism in which personality traits influence environmental factors. On the other side, socialization effects reflect the impact that environmental factors, such as major life events, have on personality.

Despite a number of studies which have identified personality plasticity across the lifespan (Lucas & Donnellan, 2011; Specht et al., 2011; Roberts et al., 2006), the relation between major life events and personality has often been ignored in previous research (cf. Roberts et al., 2008). The present paper thus seeks to investigate selection and socialization effects of one particular event in life that, with expanding globalization, has grown to become relevant for more and more young adults, namely, studying abroad.

1.3. Studying abroad and personality change

The question of who actually chooses to study abroad has been topic of only a few studies within the last decade (Luo & Jamieson-Drake, 2015; Stroud, 2010). Since previous research has not only been scarce, but also partly controversial, the first goal of this study was to examine the links between initial levels of Big Five personality traits and the commitment to studying abroad. Most findings on the relation of personality and sojourning abroad indicate that high levels of extraversion, agreeableness, conscientiousness and openness as well as low levels of neuroticism predict international mobility (Zimmermann & Neyer, 2013; Lüdtke, Roberts, Trautwein, & Nagy, 2011; Greischel, Noack, & Neyer, 2016; Jokela, 2009). Based on these previous studies, we expected to find selection effects for elevated levels of extraversion, agreeableness, conscientiousness, and openness. Meanwhile, low levels of neuroticism were also expected to predict selection to studying abroad.

Next to the examination of selection effects, there are a handful of studies that have focused on exploring the psychological effects of international experiences by comparing sojourners with stay at home students. Most findings indicate an increase in extraversion, agreeableness, and openness in consequence of studying abroad, whereas neuroticism was observed to decrease. For conscientiousness, no effect has been found (Zimmermann & Neyer, 2013; Lüdtke et al., 2011; Greischel et al., 2016). Yet again, further contributions to this body of research are required in order to generalize results. For this reason, the study's second goal was the exploration of effects of study abroad experiences on the Big Five personality dimensions. Based on the previous findings, we expected to find increases in extraversion, agreeableness and openness, and a decrease in neuroticism. For conscientiousness, we expected to find no effect (see Table 1).

2. Method

2.1. Design and procedure

This study is part of a multi-study project conducted at a small German university.[1] It features a test-retest design with two groups (sojourners vs. non-sojourners) and two measurement occasions (before and after study abroad program). Between July 2014 and February 2016, data from 370 students were collected via an online questionnaire. According to their plans for the upcoming academic semester, students were assigned to one of two groups. The first group consisted of the students who planned to attend a study abroad program in the following semester. These sojourners were approached via the international office of the university. The first measurement occasion took place approximately two to three weeks prior to the planned departure. The second measurement occasion was scheduled after a time interval of six months ($M = 6.1$ months, $SD = 1.1$), as recommended by Zimmermann and Neyer (2013), and thereby took place right after the end of the semester abroad.

For the second group, all students who did not plan on attending a study abroad program within the following semester were targeted. The recruitment of these non-sojourners was mainly organized via social networks as well as university-in-house electronic newsletters. A check for correct panel allocation was included to prevent any sojourners from taking part in the non-sojourners' survey. Analogously, two measurement occasions with a time interval of approximately six months ($M = 6.6$ months, $SD = 0.6$) were scheduled for the group of non-sojourners. As a reward for the effort, all participants had the chance to win one of three vouchers for an online bookstore and/or were granted partial course credit for participation.

2.2. Participants

2.2.1. Sample size

A total of 370 enrolled students took part in the survey at t0, 155 of which were assigned to the sojourners group, whereas 215 were non-sojourners. At this point, we excluded data from six students because these participants completed the questionnaire twice. At t1 data were obtained from a total of 243 students (96 sojourners and 147 non-sojourners). Again, 17 cases had to be excluded for the same reason as before. A subsequent analysis revealed that data from another five participants who took part at t1 needed to be excluded from further analyses due to the impossibility to match the data sets to those of t0. Hence, the total sample size cleaned from irrelevant data added up to 221 participants, with 93 data sets of sojourners and 128 data sets of non-sojourners taken into account.

2.2.2. Basic characteristics

The age of participants at t0 ranged from 18 to 48 years (sojourners: $M = 22.31$, $SD = 2.03$; non-sojourners: $M = 22.84$, $SD = 4.28$), with 90% of all participants being aged approximately 19 to 29 years. With a share of 83.3% (sojourners: 81.7%; non-sojourners: 84.4%), distinctly more women than men took part in the study, $\chi^2(1, N = 221) = 549.69$, $p < 0.001$. Considering the sex ratio in the given population of the university students in general (60.0% female, $N = 9076$; Leuphana University of Lüneburg, 2014) and in the given population of students accepted into study abroad programs by the international office (64.0% female, $N = 553$; personal communication, March 23, 2016), analyses reveal strong effects for gender on participation in the study, especially for the non-sojourners, $\chi^2(1, N = 128) = 18.73$, $p < 0.001$, but also for the sojourners, $\chi^2(1, N = 93) = 7.00$, $p < 0.010$. Among both sojourners and non-sojourners, 46.2% of students had previous international experiences, however, the share among sojourners was considerably higher with 53.8% versus 40.6% among non-sojourners, $\chi^2(1, N = 221) = 3.74$, $p = 0.053$.[2]

2.2.3. Attrition effects

To test for attrition effects, students who participated at both measurement occasions ("continuers") were compared with students who only completed the first questionnaire ("dropouts"). A total of 143 participants dropped out prior to t1, whereas 221 participants took part at

[1] As part of this multi-study project, effects of studying abroad on self-efficacy were also studied, and a corresponding article has recently been published (Petersdotter, Niehoff, & Freund, 2017).

[2] More detailed information on basic characteristics of the study's sample can be found in the Online Appendix (A-1).

Table 1
Overview of hypotheses.

Selection hypotheses		Socialization hypotheses	
H1	High extraversion predicts selection to study abroad.[a,b,c,e]	H6	Studying abroad positively affects extraversion.[a]
H2	High agreeableness predicts selection to study abroad.[a,e]	H7	Studying abroad positively affects agreeableness.[a,e]
H3	High conscientiousness predicts selection to study abroad.[b]	H8	Studying abroad has no effect on conscientiousness.[a,b]
H4	Low neuroticism predicts selection to study abroad.[a,e]	H9	Studying abroad negatively affects neuroticism.[a,b,d]
H5	High openness predicts selection to study abroad.[a,c]	H10	Studying abroad positively affects openness.[b,e]

[a] Lüdtke et al., 2011.
[b] Zimmermann & Neyer, 2013.
[c] Jokela, 2009.
[d] Ying, 2002.
[e] Greischel et al., 2016.

both occasions. The return rate therefore constitutes 60.7% (sojourners: 61.6%; non-sojourners: 60.1%). With respect to the Big Five, continuers and dropouts were very alike on most dimensions. Nevertheless, continuers were significantly more conscientious than dropouts (continuers: $M = 3.78$, SD $= 0.62$; dropouts: $M = 3.60$, $SD = 0.60$; $p < 0.010$, $d = 0.29$) – an effect that has already been reported in various other studies (Lüdtke et al., 2011; Specht et al., 2011).

2.3. Measures

2.3.1. Big Five

The Big Five personality dimensions were assessed using a German version of the Big Five Inventory (BFI; for more information see Lang, Lüdtke, & Asendorpf, 2001). Table 2 displays mean averages, retest-reliabilities and internal consistencies for all scales reported in a norm sample of young adults (Lang et al., 2001) as well as observed in the present study. Both reported retest-reliabilities and internal consistencies range within an acceptable to good spectrum, with the exception of agreeableness ($r_{tt} = 0.55$, $\alpha = 0.70$). For the present sample, almost all scales showed even higher reliability scores than reported by Lang and colleagues. However, CFAs for each factor and at each measurement occasion revealed relatively poor fit in most cases, so that all following analyses were conducted using manifest variables.[3]

2.3.2. Additional measures

In addition to the Big Five, the online questionnaire obtained basic demographic information such as age, gender, enrollment status, and major study subject. Furthermore, all students were asked to indicate previous international experiences (without parental company, for at least three months) and rate a number of arguments, stating in how far they personally considered these arguments as reasons to either study abroad or stay at home. Given arguments were, for example, financial reasons, lacking interest, or the intention to go abroad at a later point in their educational career.

2.4. Analytical strategy

We conducted a hierarchical binary logistic regression in order to test for self-selection effects.[4] Study abroad served as the dependent variable. In Model I, we controlled for effects of age, gender and previous international experiences. In Model II, mean averages for Big Five personality dimensions at t0 were added as predictors. Only non-sojourners who did not indicate the intention to study abroad at a later point of their college education were included in these analyses. This decision was based on the assumption that Big Five selection effects would merely occur for the intention to study abroad rather than the actual sojourn, thereby controlling for reasons other than personality that might have influenced the decision not to study abroad.

For the analysis of socialization effects, a series of multivariate regression models with the Big Five measured at t1 as dependent variables was set up. In Model I, Big Five ratings at t0, age, and gender were included as independent variables. In Model II, the dichotomously coded variable study abroad (0 = no, 1 = yes) was added. Finally, an interaction term (study abroad × t0 Big Five rating) was added in Model III. For directionally formulated hypotheses, p-values were halved.

3. Results

3.1. Selection effects

The hierarchical binary logistic regression revealed that commitment to a study abroad program was significantly predicted by age, $Exp(B) = 0.834$, $p = 0.003$ and previous international experiences, $Exp(B) = 2.002$, $p = 0.038$. This means that younger students as well as students who already had international experiences were more likely to study abroad. With an R^2 of 0.118, $p = 0.002$ this model containing only control variables performed significantly better than the null model.

Adding personality ratings at t0 increased the pseudo R^2 to 0.176, $p = 0.003$. Yet, the inclusion did not significantly improve model fit, as demonstrated by an insignificant chi-square change, $\Delta\chi^2 = 8.053$, $p = 0.153$. For agreeableness, an odds ratio of $Exp(B) = 1.930$, $p = 0.033$ was observed. Moreover, the expected effect for openness was affirmed, $Exp(B) = 1.644$, $p = 0.038$. In addition, the control variable gender also became significant with $Exp(B) = 3.121$, $p = 0.032$. Thereby, students high in agreeableness and/or high in openness and male students were more likely to study abroad.

3.2. Socialization effects

Table 3 shows the results of the multivariate analyses predicting t1 Big Five ratings. Unsurprisingly, Model I showed highly significant effects for t0 Big Five ratings, which are just the test-retest stability coefficients while taking into account the covariates of age and gender. For extraversion, the results of Model II suggest that studying abroad has a significant, positive impact on t1 extraversion, indicating that students who studied abroad increased in extraversion over time. However, this significant main effect disappeared when the interaction between t0 ratings and study abroad was taken into consideration. Fig. 1 plots the mean levels of Big Five ratings at t0 and t1 for sojourners and non-sojourners to illustrate the interaction effects.

Study abroad was a significant and positive predictor for t1 agreeableness ratings. Accordingly, sojourners increased in agreeableness when studying abroad. This main effect remained significant when the interaction between t0 ratings and study abroad was taken into account. The interaction term negatively predicted t1 agreeableness ratings. This negative interaction effect indicates that students who were initially lower in agreeableness benefited more from the positive impact of study abroad on t1 agreeableness ratings than those who were highly agreeable to begin with. In order to visualize this effect we compared

[3] Fit indices for CFA models are included in Appendix A-2.
[4] Zero-order correlations between all variables in the regression models are listed in Appendices A-3 and A-4.

Table 2
Big Five reported (Lang et al., 2001) and observed mean averages, internal consistencies and retest-reliabilities.

	Lang et al. (2001)				Observed																			
	Young adults ($N = 120$)				Sojourners ($N = 93$)								Non-sojourners ($N = 128$)								Non-Sojourners w/o intention ($N = 73$)					
					t0				t1				t0				t1				t0				t1			
Personality trait	*M*	*SD*	α	r_{tt}	*M*	*SD*	α	r_{tt}	M	SD	α	r_{tt}	*M*	*SD*	α	r_{tt}	*M*	*SD*	α	r_{tt}	*M*	*SD*	α	r_{tt}	*M*	*SD*	α	r_{tt}
Extraversion	3.53	0.68	0.82	0.78	3.48	0.65	0.86	0.80	3.59	0.68	0.86	0.80	3.54	0.80	0.89	3.51	0.77	0.88	0.88	3.48	0.82	0.89	3.46	0.81	0.89	0.86
Agreeableness	3.50	0.52	0.67	0.55	3.84	0.48	0.65	0.73	3.89	0.45	0.64	0.73	3.71	0.57	0.75	3.66	0.62	0.78	0.86	3.66	0.60	0.76	3.63	0.62	0.79	0.87
Conscientiousness	3.62	0.52	0.75	0.69	3.81	0.60	0.80	0.73	3.80	0.54	0.76	0.73	3.75	0.64	0.82	3.69	0.68	0.84	0.89	3.74	0.66	0.82	3.70	0.68	0.83	0.88
Neuroticism	2.65	0.60	0.77	0.69	2.81	0.76	0.82	0.75	2.75	0.73	0.81	0.75	2.74	0.70	0.80	2.86	0.74	0.82	0.80	2.78	0.72	0.79	2.91	0.79	0.84	0.81
Openness	3.55	0.52	0.78	0.82	3.58	0.61	0.80	0.85	3.59	0.62	0.82	0.85	3.48	0.62	0.79	3.48	0.63	0.80	0.87	3.39	0.66	0.80	3.40	0.68	0.82	0.88

Note: *M* is the mean average, *SD* the standard deviation of the mean average. α is Cronbach's alpha for the internal consistency of the trait's scale. r_{tt} is the retest-reliability of the mean average. A 5-point rating scale (1–5) was employed. "Non-Sojourners w/o intention" are students who stayed at home without the intention to study abroad at a later point of their academic career.

the mean averages of the 25% initially least agreeable sojourners (t0: $M = 3.25$, $SD = 0.26$; t1: $M = 3.54$, $SD = 0.47$) with those of the 25% initially most agreeable sojourners (t0: $M = 4.27$, $SD = 0.28$; t1: $M = 4.22$, $SD = 0.30$). Whereas the least agreeable sojourners increased about a whole standard deviation ($d = 0.764$, $p < 0.001$), the sojourners who were most agreeable prior to their departure showed only little and non-significant deviation ($d = -0.172$, $p = 0.242$).

Conscientiousness seemed to remain largely unaffected by study abroad. However, when taking into account the interaction between t0 ratings and study abroad, study abroad became a highly significant positive predictor for t1 conscientiousness ratings. This means that students who studied abroad increased in conscientiousness over time. Moreover, the interaction term was a highly significant negative predictor for t1 conscientiousness ratings. Again, the negative interaction term suggests that students who were initially lower in conscientiousness got more conscientious as a consequence of their study abroad experience, while students who were initially highly conscientious decreased slightly in conscientiousness over time. We again compared mean averages of the 25% initially least conscientious sojourners (t0: $M = 3.06$, $SD = 0.34$; t1: $M = 3.32$, $SD = 0.28$) and the 25% initially most conscientious sojourners (t0: $M = 4.41$, $SD = 0.27$; t1: $M = 4.20$, $SD = 0.49$). Both groups experienced substantial changes about almost a whole standard deviation in conscientiousness. However, while the level increased in the group of less conscientious students ($d = 0.835$, $p < 0.001$), it decreased for the initially highly conscientious students ($d = -0.531$, $p < 0.001$).

Studying abroad had a significant and negative impact on t1 neuroticism, indicating that sojourners decreased in neuroticism when studying abroad. However, this significant main effect disappeared when the interaction between t0 ratings and study abroad was added to the model. Yet, t1 ratings for neuroticism were significantly and positively predicted by the control variable age. This means that older sojourners rated higher in t1 neuroticism levels than younger sojourners.

T1 ratings for openness were unaffected by study abroad and remained unaffected when the interaction between t0 ratings and study abroad was added. This means that sojourners did not significantly change in openness when studying abroad.

4. Discussion

4.1. Selection effects

The present data revealed that participants with high levels of agreeableness and openness to experience were more likely to study abroad. Regarding neuroticism, extraversion, and conscientiousness, we did not find significant results. With more and more employers appreciating or even requesting international experiences in applicants, it can be assumed that an increasing number of students choose to study abroad out of external reasons instead of internal motivation (Deutscher Akademischer Austauschdienst, 2011). With this development, personality differences between sojourners and non-sojourners might be decreasing, resulting in a lack of observed selection effects.

While the identified relation for gender can be explained by the above mentioned gender effect on participation, the observed effect of age was unexpected. Due to the increasing pressure to finish college education as quickly as possible (Gomez Tutor, Hobelsberger, & Menzer, 2011), fear of an extension of the prescribed period of study might be the reason why older students could be opposed to studying abroad (Deutscher Akademischer Austauschdienst, 2011). The observed positive effect of previous international experience on commitment to studying abroad might be linked to the edge of experience that previous sojourners have: because they know what to expect, pervious sojourners are possibly less threatened by the idea of living abroad and hence more likely to commit to a study abroad program. We would like to advise future researchers to take students' motivation for

Table 3
Effects of study abroad on the Big Five: multivariate analyses results.

Predictors	t1 extraversion			t1 agreeableness			t1 conscientiousness			t1 neuroticism			t1 openness		
	R^2	*B*	*SE*	R^2	*B*	*SE*	R^2	*B*	*SE*	R^2	*B*	*SE*	R^2	*B*	*SE*
Model I	0.714**			0.667**			0.695**			0.589**			0.751**		
Age		−0.002	0.008		−0.012	0.006		−0.010	0.007		0.020*	0.009		−0.010	0.006
Gender		−0.041	0.072		−0.012	0.061		−0.049	0.065		−0.105	0.088		0.007	0.059
t0 rating		0.823**	0.032		0.834**	0.040		0.821**	0.037		0.767**	0.041		0.884**	0.034
Model II	0.721**			0.681**			0.697**			0.602**			0.752**		
Age		0.000	0.008		−0.010	0.006		−0.009	0.007		0.018	0.009		−0.010	0.006
Gender		−0.051	0.071		−0.024	0.060		−0.053	0.065		−0.090	0.087		0.005	0.059
t0 rating		0.825**	0.032		0.825**	0.040		0.820**	0.037		0.774**	0.040		0.885**	0.034
Study abroad		0.126**	0.052		0.124**	0.044		0.048	0.047		−0.061**	0.063		0.020	0.043
Model III	0.723**			0.689**			0.717**			0.604**			0.753**		
Age		0.000	0.008		−0.009	0.006		−0.006	0.007		0.018*	0.009		−0.010	0.006
Gender		−0.049	0.071		−0.025	0.059		−0.089	0.064		−0.099	0.087		0.006	0.060
t0 rating		0.840**	0.039		0.888**	0.047		0.939**	0.045		0.810**	0.054		0.884**	0.045
Study abroad		0.254	0.242		0.859**	0.314		1.230**	0.282		−0.072	0.231		−0.014	0.251
t0 × study abroad		−0.037	0.068		−0.193*	0.082		−0.311**	0.073		−0.084	0.080		0.009	0.070

Note. $N = 221$. Gender was coded as 0 = female, 1 = male. Study abroad was coded as 0 = no study abroad, 1 = study abroad. *B* estimates are unstandardized coefficients.
* $p < 0.05$.
** $p < 0.01$.

studying abroad and their reasons to stay at home into account when investigating selection effects.

4.2. Socialization effects

At first glance, most hypotheses regarding socialization effects of study abroad programs seem to be confirmed. Sojourning affects extraversion and agreeableness positively. Neuroticism decreases as a result of study abroad experiences, whereas conscientiousness remains unchanged. Yet, the results of Model II also revealed two unexpected findings, which will be discussed in the following.

First, openness remained unaffected by study abroad, which goes against all extant research. Although levels of openness were not particularly high compared to the norm sample by Lang et al. (2001), the high share of cultural studies majors among the sojourners might have provoked this finding. For cultural studies majors, who actively engage with cultural and artistic topics in their studies, a study abroad experience might be used to focus on other aspects of student life, such as social interactions – as can be seen in the increasing level of agreeableness. The factor openness, however, might not be influenced as strongly due to a (relative) reduction of engagement with artistic topics during their stay abroad. Moreover, students might be confronted with their own prejudices or ignorance of other cultures when faced with a foreign culture, and therefore lower the perception of their own level of openness. Martin, Katz-Buonincontro, and Livert (2015) proposed that social comparison might also play a role. When sojourners get to know international students from around the world, they might change who they compare to when rating their own level of openness – which could

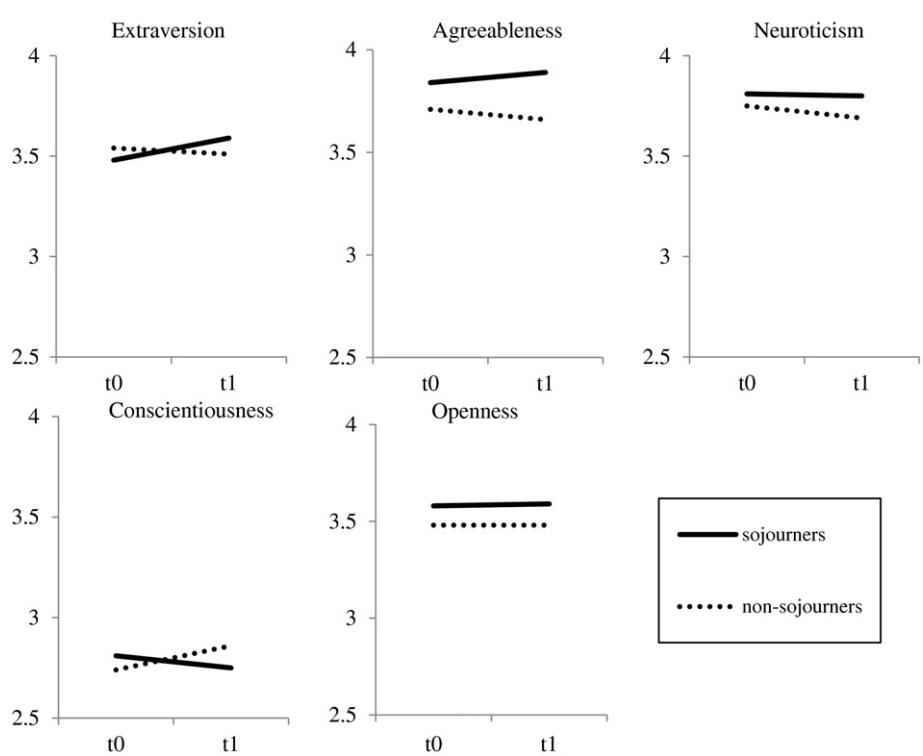

Fig. 1. Sojourners' mean levels of Big Five before studying abroad (t0) and after studying abroad (t1) compared to non-sojourners' mean levels.

lead to lower ratings than what might be appropriate (Martin et al., 2015).

The model for neuroticism revealed the second surprising finding: t1 ratings for neuroticism were not only negatively influenced by studying abroad, but also positively predicted by age. Hence, older students rated higher in neuroticism at the second measurement occasion than younger students which is surprising given the distinct research on general maturation processes that reports decreases in neuroticism with age (Roberts et al., 2006). Again, what might account for this observation is the pressure to finish college education as quickly as possible (Gomez Tutor et al., 2011). Over time, especially older students might increasingly be stressed or feel pressured to finish education, thus evoking increases in neuroticism. Yet again, further research needs to be conducted in order to find evidence for supporting or rejecting this possible explanation.

Except for these unexpected findings regarding openness and the relation of age and neuroticism, all other observed effects were in line with previous empirical reports. However, these effects could only be observed in a model that did not consider interactions between t0 ratings and study abroad. Upon inclusion of the interaction terms, the situation was much different: sojourning only affected agreeableness and conscientiousness positively, thus, all socialization hypotheses except for agreeableness had to be rejected. We argue that in order to truly identify the effects of studying abroad on the Big Five, it is not only advisable to control for initial personality scores but also to add interactions between initial score and study abroad status. Extant research regarding the effects of studying abroad has mainly focused on mean-level changes, and has hence seemingly ignored potential interaction effects so far. However, note that the standard errors of the estimates for study abroad are considerably larger when including the interaction terms. Since the interaction terms are products of t0 ratings and the dichotomously coded study abroad variable, they highly correlate with study abroad ($r > 0.9$).

The significant negative interaction terms in both models indicated that initial levels of agreeableness and conscientiousness moderate the positive relation between study abroad and t1 ratings, suggesting that students with low initial scores benefit from studying abroad. Yet, especially regarding conscientiousness, one might argue that also students with high initial scores, who decrease in levels over time, benefit from international experiences: a comparison of sojourners' conscientiousness ratings upon return with those of the norm sample reveals that mean-levels are significantly closer to the conscientiousness level reported by Lang et al. (2001) after studying abroad. Hence, a study abroad experience affects conscientiousness in the way that it makes sojourners less extreme.

4.3. Relevance and limitations

In general, the findings of the present study indicate that studying abroad can be characterized as a major life event with the ability to provoke personality changes. We could identify selection as well as socialization effects for the Big Five that shed more light on the ongoing discussion about the impact of major life events on personality.

Still, the present study has several limitations. First, and most importantly, the design holds restrictions, since it is non-experimental and therefore cannot guarantee that the observed effects did not occur due to other uncontrolled variables. Moreover, the possibility of common-method bias needs to be considered, since the personality ratings were assessed via self-reports that are open to cognitive biases. The CFAs testing the psychometric properties of the Big Five scales showed deficiencies so that latent variable models could not be analyzed. While this is a problem that often arises when standard psychometric measures are used in a specific study, it limits the interpretability of the findings because the analytical methods can only employ the manifest variables, thus omitting measurement error. Furthermore, the results of Model III have to be viewed in consideration of multicollinearity among the predictor variables.

Above that, there are limitations related to the study's sample. First, strong selection effects of gender for participating in the study could be identified in so far that women participated more. Second, the generalizability of the results is limited due to the high homogeneity of the sample, since all students were recruited from the same university. Third, the sample size of a total of 221 students only allowed for the detection of comparatively large effects. Lastly, all observed effects are only valid for sojourners who stayed abroad for the whole duration of their sojourn, since no cases of program dropouts occurred.

Finally, the study did not identify mediating mechanisms that account for the observed relations. In order to successfully explore the relationship of studying abroad and the Big Five personality dimensions, future research should focus on identifying the socio-psychological processes that explain the ways in which studying abroad shapes personality. Additionally, details of the nature of the sojourn, such as the country of destination, should be included in the analyses, as proposed and modeled by Ward, Wilson, and Fischer (2011).

Yet, the present study went beyond previous research by identifying substantial interaction effects for initial levels of agreeableness and conscientiousness and studying abroad. Especially for conscientiousness, these findings are interesting, since all previous studies proposed that international sojourning had no impact on conscientiousness. However, previous research has mainly focused on main effects, thereby ignoring the strong moderation effect for initial personality levels that could be observed in the present study. Future research should pay major attention to this moderation, since it might be able to account for a variety of observations that could not yet be successfully explained.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.paid.2017.02.043.

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